Mid-Term Lab Examination — 8-bit Accumulator

University of Asia Pacific

Department of Computer Science and Engineering

Course Code : CSE 450 (VLSI Lab)

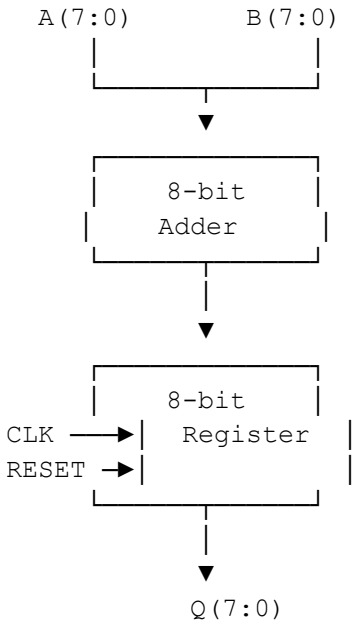
Mid-Term Lab Examination | Group C1 | Spring 2026Duration: 1 hr Total Marks: 20

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Mid-Term Task: 8-bit Accumulator

Objective. Design and implement an 8-bit accumulator in VHDL by combining the 8-bit adder and register concepts developed in the laboratory. The system must perform addition synchronously with the clock and store the result in an 8-bit register.

System Architecture



Functional requirement.

At every rising edge of CLK, the register stores the sum of the two 8-bit inputs. RESET has priority over the addition operation.

Required Operation

RESET	CLK	A, B	Operation / Q
1	↑	XX, XX	Q ← 00000000
0	↑	A, B	Q ← A + B
0	No rising edge	XX, XX	Q remains unchanged

Required Test Sequence

Your testbench must produce the following sequence. The expected Q value should be observed after the corresponding rising clock edge.

Step	RESET	A	B	Operation	Expected Q / Explanation
1	1	00000000	00000000	Reset	00000000 — Register is cleared
2	0	00000011	00000101	3 + 5	00001000 — Load sum
3	0	00000010	00000001	2 + 1	00000011 — Load new sum
4	0	11111111	00000001	255 + 1	00000000* — 8-bit overflow
5	0	10101010	00000101	170 + 5	10101111 — Load new sum

* Note: For Step 4, the mathematical result is 256, but an 8-bit output can represent only 0–255. Therefore, Q = 00000000 and the carry-out is discarded unless your implementation explicitly exposes it.

Implementation Requirements

• Use VHDL and IEEE.STD_LOGIC_1164.ALL.

• The register operation must be synchronized with the rising edge of CLK.

• RESET must be synchronous and active-high.

• Do not use a behavioral integer-based accumulator as a replacement for the required adder/register structure.

• The design should be modular and clearly identify the 8-bit adder and 8-bit register portions.

• The submitted testbench must generate a clock and apply the required input sequence.

Submission

While submitting to the classroom prepare the report from the third page of this template including the current template in the first and second page. From the third page write your report including but not limited to the following points.

• Attach your entity code and testbench code

• Attach your output visuals

• Challenges faced during the experiment

• Conclusion

MID-TERM LAB EXAMINATION REPORT

8-bit Accumulator

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1. Objective

The objective of this experiment was to design and implement an 8-bit accumulator in VHDL. The accumulator combines an 8-bit combinational adder with an 8-bit register so that the addition result is stored synchronously on the rising edge of the clock. The design also uses a synchronous, active-high reset.

2. Design and Implementation

The design was implemented in a modular structural manner using VHDL and IEEE.STD_LOGIC_1164.ALL. The 8-bit adder was constructed from lower-level adder components, and the 8-bit register stores the adder output. At the top level, inputs A and B are applied to the 8-bit adder, whose SUM output is connected to the D input of the 8-bit register. CLK and RESET control the register, and the register output is Q.

3. System Architecture

The final data path is:

A(7:0), B(7:0) → 8-bit Adder → SUM(7:0) → 8-bit Register → Q(7:0)

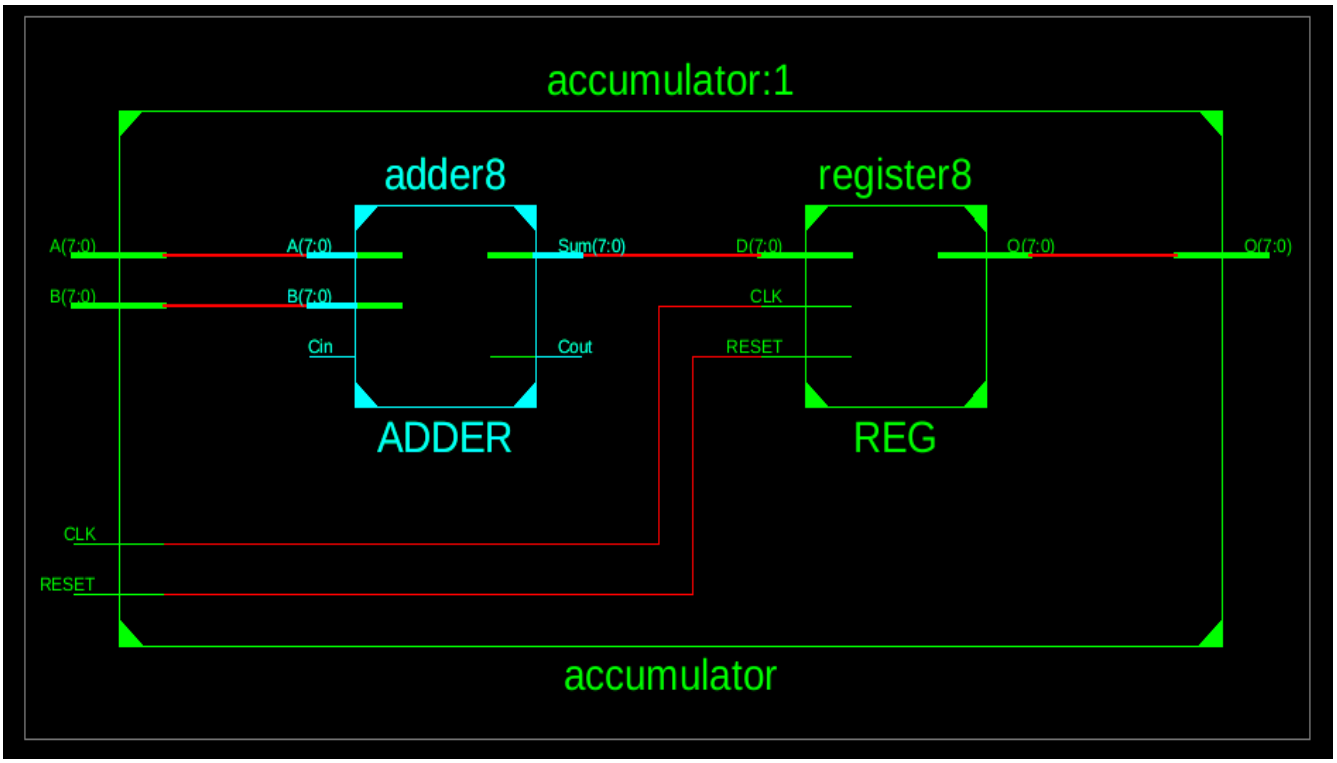
4. Important Test Cases

Step	RESET	A	B	Expected Q
1	1	00000000	00000000	00000000
2	0	00000011	00000101	00001000
3	0	00000010	00000001	00000011
4	0	11111111	00000001	00000000
5	0	10101010	00000101	10101111

The fourth case demonstrates 8-bit overflow: 11111111 + 00000001 produces a mathematical result of 256, so the 8-bit stored output is 00000000 and the carry-out is discarded.

5. Output Visuals

Figure 1. ISE 14.7 Project Hierarchy



Insert a screenshot showing the completed VHDL source hierarchy, including the adder, register, accumulator, and testbench.

Figure 2. Accumulator Simulation Waveform



Insert the most important ISim waveform screenshot showing CLK, RESET, A, B, and Q for the required test sequence.

6. Challenges Faced

The primary implementation challenge was maintaining accurate structural connections between the lower-level adder components and the final 8-bit modules. Particular attention was required when connecting carry signals between the different adder stages and linking the adder's SUM output to the register input. Another important consideration was ensuring that the RESET signal operates synchronously with the rising edge of the CLK signal.

7. Conclusion

An 8-bit accumulator was successfully designed and implemented in VHDL by integrating an 8-bit adder with an 8-bit register. The design stores the calculated addition result on the rising edge of the clock and clears the register when the synchronous active-high RESET signal is asserted. The test sequence verifies normal addition, consecutive accumulator updates, and 8-bit overflow behavior. The modular structural design also simplifies simulation, verification, and debugging in Xilinx ISE 14.7.

8. Github

https://github.com/Tasnim3322Abrar/ISE-14.7-for-VLSI